

Gesture Responsive Arm for Intuitive Linkage (GRAIL)

Group Members:

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Project Idea



Many real-world tasks require precise manipulation in environments where direct human access is difficult, dull, or dangerous. Existing solutions rely on expensive, pre-programmed robotic systems that lack intuitive real-time human control.

This project presents a low-cost, vision-based gesture-controlled robotic hand that mirrors finger movements in real time using just a webcam and an ESP32 — no wearables, no complex controllers. Unlike industrial systems costing thousands of dollars, this approach is lightweight, portable, and requires no physical contact with the operator, making it viable for surgery assistance, bomb disposal, etc.

Project Approach



Design

- Problem definition
- Block Diagram
- Mechanical design
- Component selection

Prototyping

- 3D printing of arm components
- Assembly of joints and structure
- Circuit setup
- Initial firmware development
- First movement tests

Integration

- Mechanical + electronics integration
- Motor calibration
- PWM control implementation
- Software refinement
- System synchronization

Testing & Validation

- Movement accuracy testing
- Reliability testing
- Debugging and improvements

Project Overview



System Description

The system uses a USB camera and MediaPipe on a PC to track hand gestures, transmitting data via UART to an ESP32, which drives six servo motors to replicate the movements on a robotic hand.

Degrees of Freedom (DOF)

- 5 DOF – Individual finger actuation
- +1 DOF – Forearm

Overall Outcome

At the end of the project, the system will demonstrate a human-in-the-loop gesture-controlled robotic arm where finger, and forearm movements of the user are reflected in real time by the robotic arm with smooth and stable motion.

Bill of Materials (BOM)



Project: GRAIL
Total Cost : 4,247/-

Serial No.	Category	Part	Part Name	Qty	Unit Cost	Cost	Link
1	▼ Microcontroller	ESP32 Wroom	ESP32 Wroom 32D	1	₹369.00	₹369.00	link
2	▼ Motors	SG90	Servo Motor	1	₹142.00	₹142.00	link
3	▼ Motors	MG995	Servo Motor	5	₹240.00	₹1,200.00	link
4	▼ Frame	3D printed	NA	1	₹2,536.00	₹2,536.00	link
				8	Total Cost	₹4,247.00	

[Links](#)

Why choose GRAIL?:



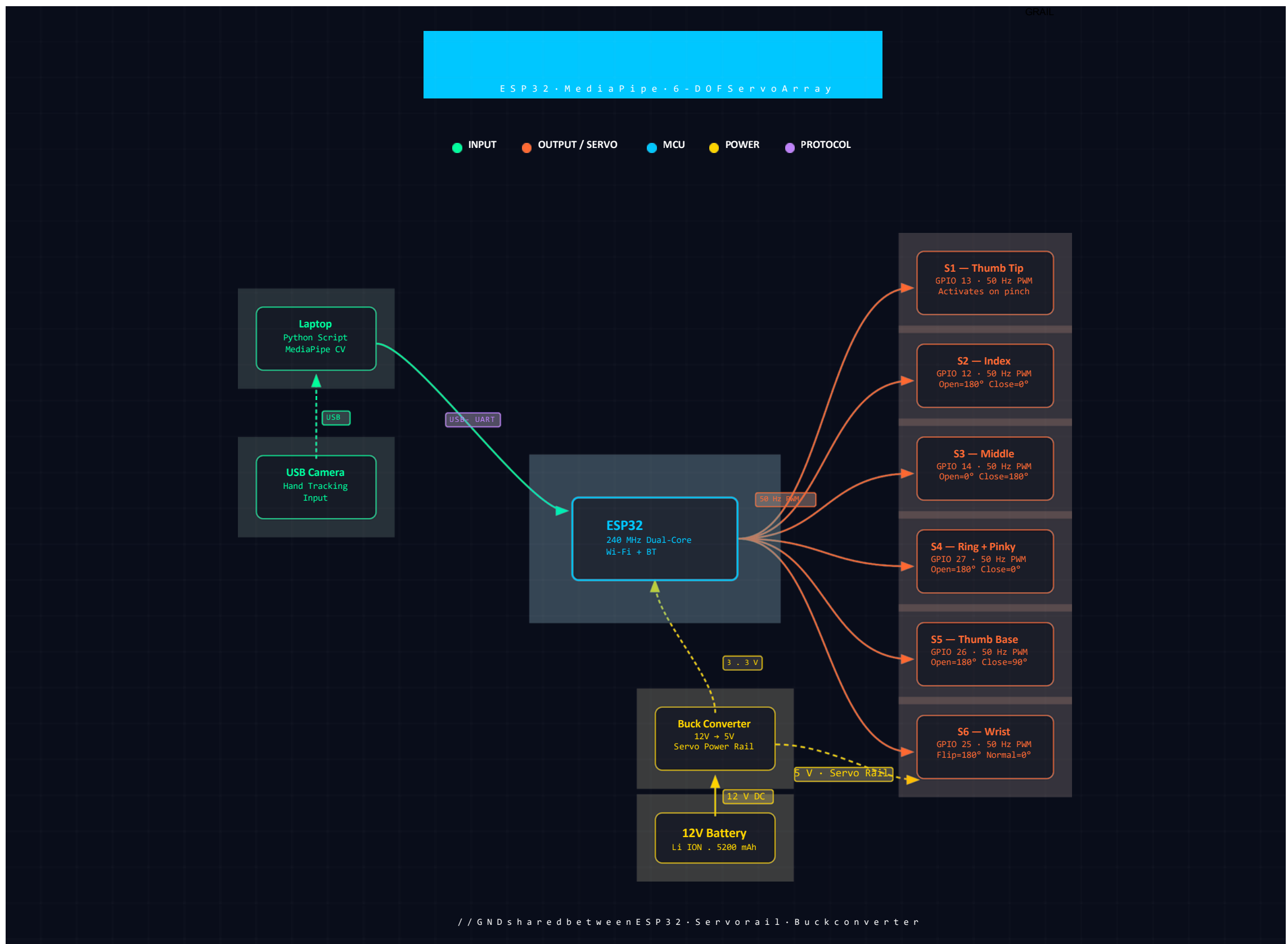
Where it helps society:

- Bomb disposal & hazardous environments — Global EOD robot market is \$1.2 billion (2023), projected \$2.1 billion by 2030. Most units cost \$50,000–\$150,000. A vision-based low-cost arm attachment could democratize access for smaller defense forces and NGOs.
- Remote surgery assistance in rural India. A low-cost teleoperation arm for basic procedures could bridge this gap where da Vinci is economically impossible.
- Rehabilitation — Global rehabilitation robotics market: \$2.6 billion (2023), growing at 18% CAGR. Current devices cost \$20,000–\$80,000 per unit. GRAIL's vision-based approach needs no physical attachment to the patient's hand.
- Industrial automation for SMEs — 90% of Indian manufacturing is SMEs. Less than 3% use any form of robotic automation (IBEF 2023) due to cost. A sub-₹10,000 gesture-controlled arm assistant is viable at this scale.
- Education & Research — An open, low-cost, fully documented gesture-controlled robotic platform for universities and labs.

GRAIL's position:

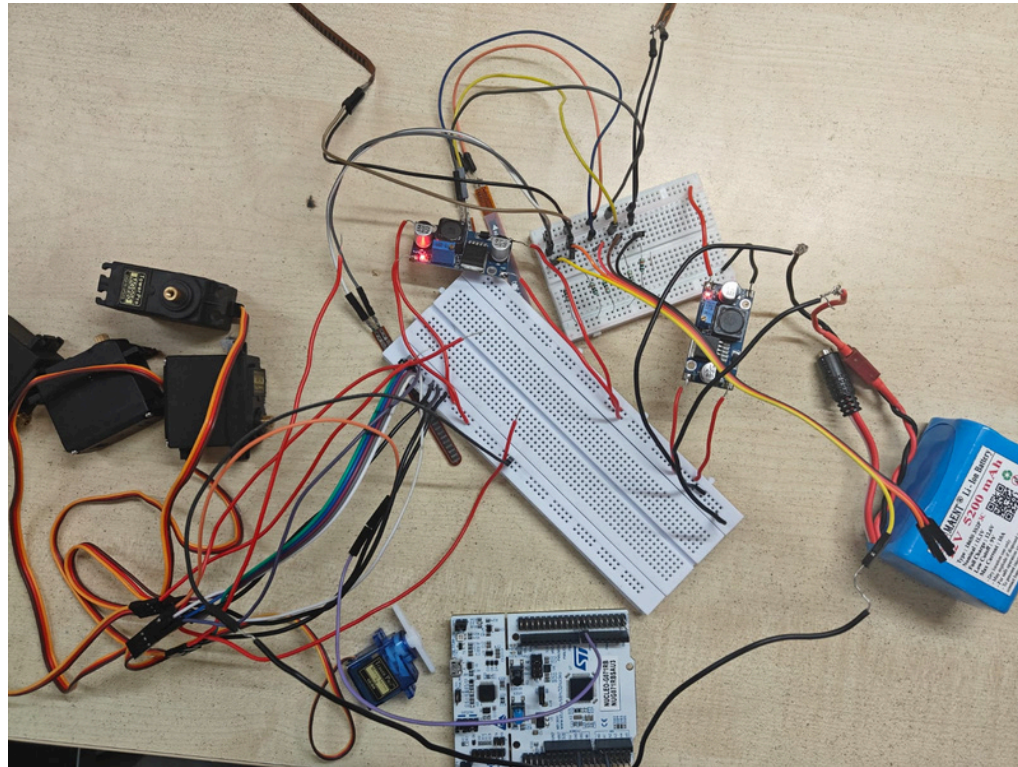
- Estimated BOM cost: under ₹5,000 (\$60)
- No wearables, no contact with operator
- Works with any standard webcam
- Deployable on any laptop — no dedicated hardware
- Extensible to custom gestures via the trained model

Block Diagram

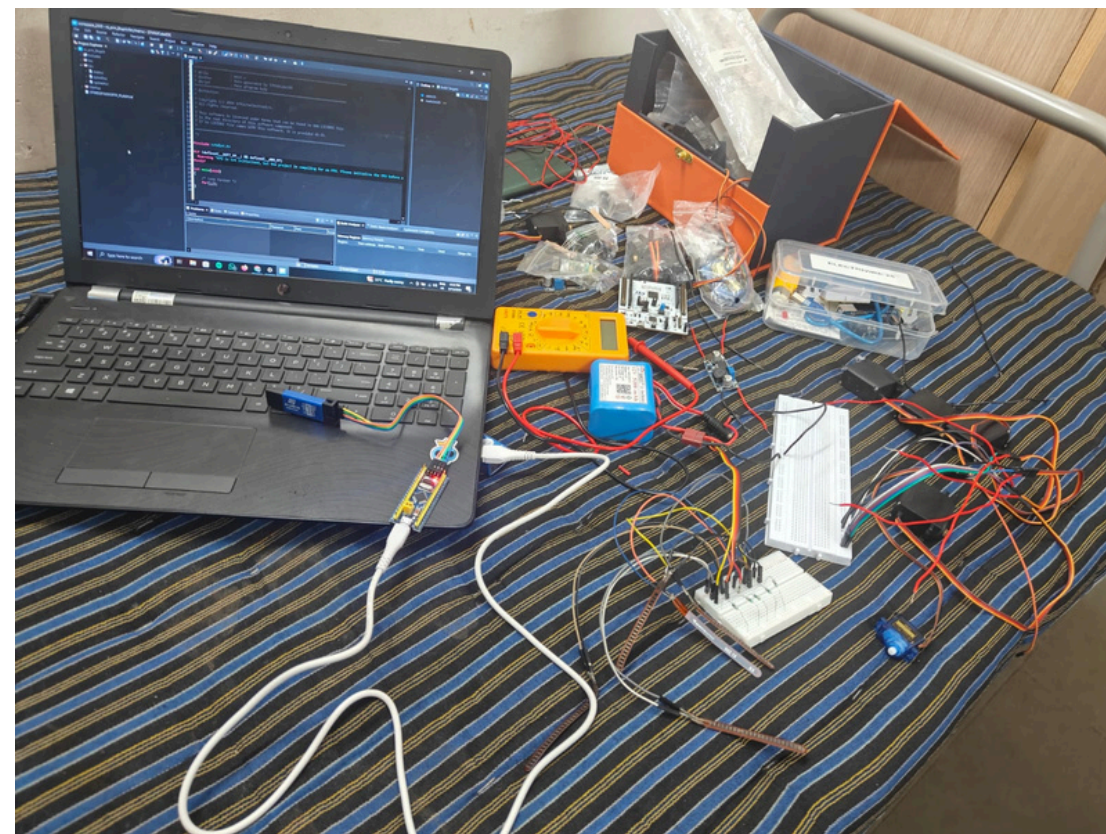


Block Diagram:
[Link](#)

Phase -1:

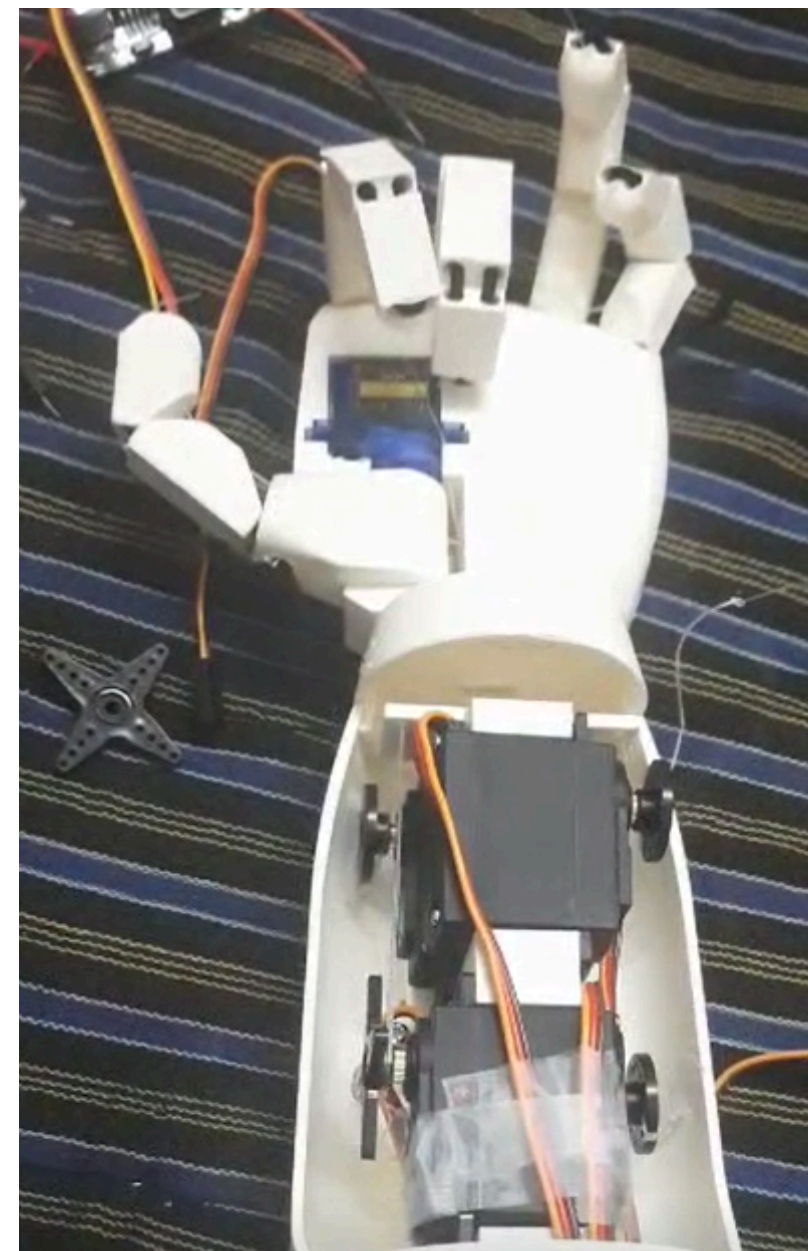
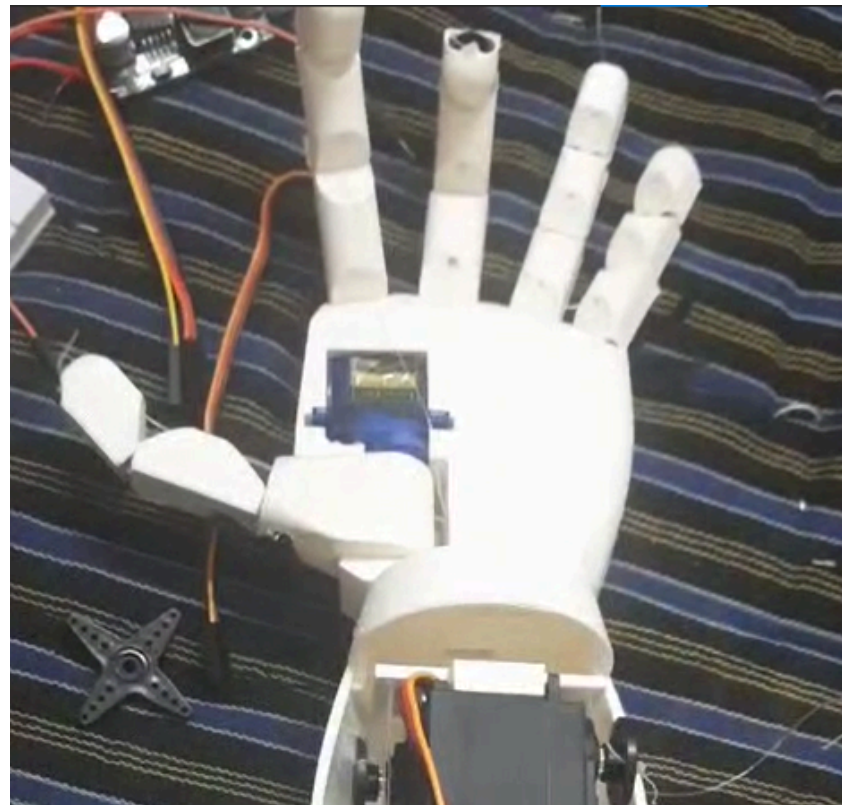


[Block Diagram](#)



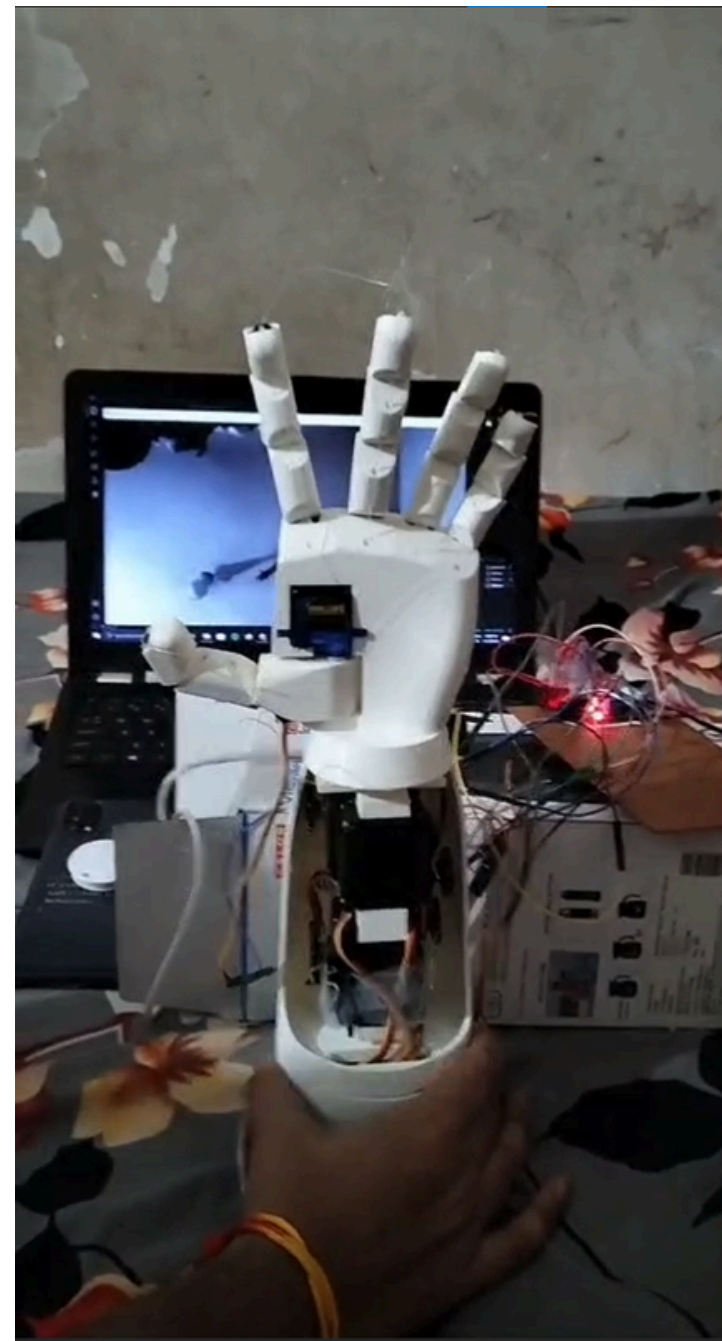
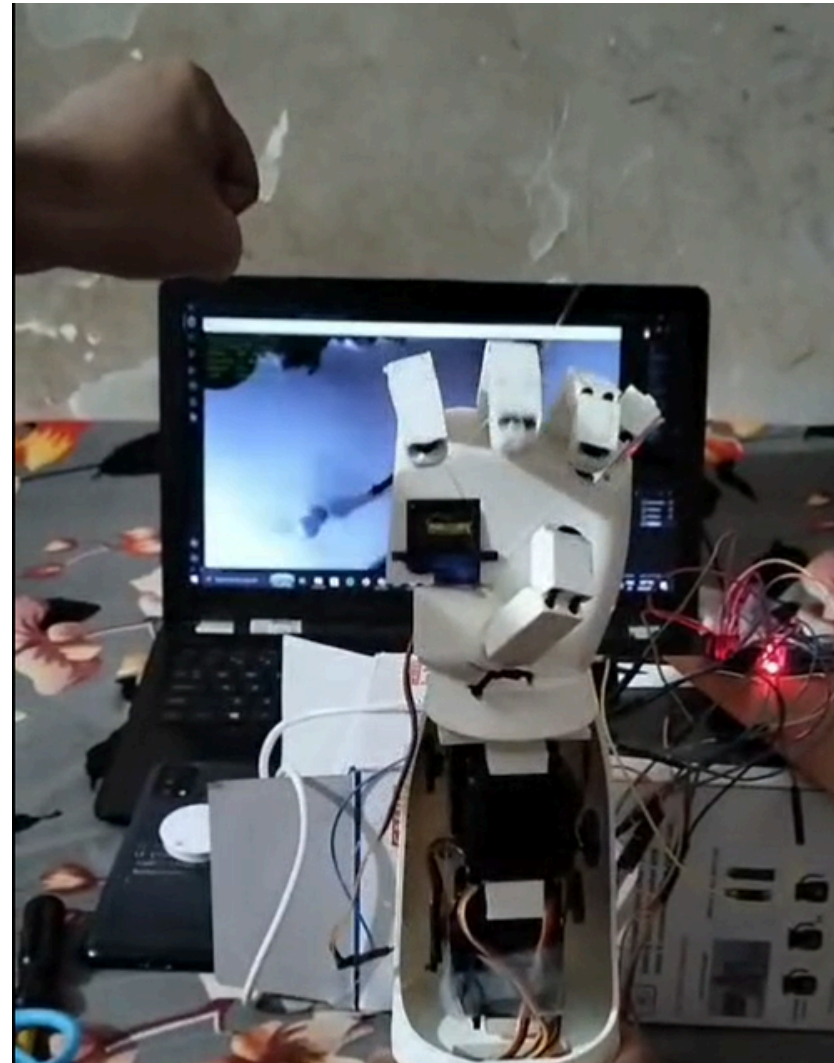
Designed a wearable glove with 5 flex sensors and an IMU for wrist tracking. Two STM32 boards (Nucleo + Bluepill) were planned — one for the glove, one for the robotic hand — communicating via Bluetooth modules. Stopped because the STM32 Nucleo was physically damaged in the lab and the Bluepill was identified as a counterfeit clone with unstable flash and USB behavior, making reliable development impossik

Phase -2:



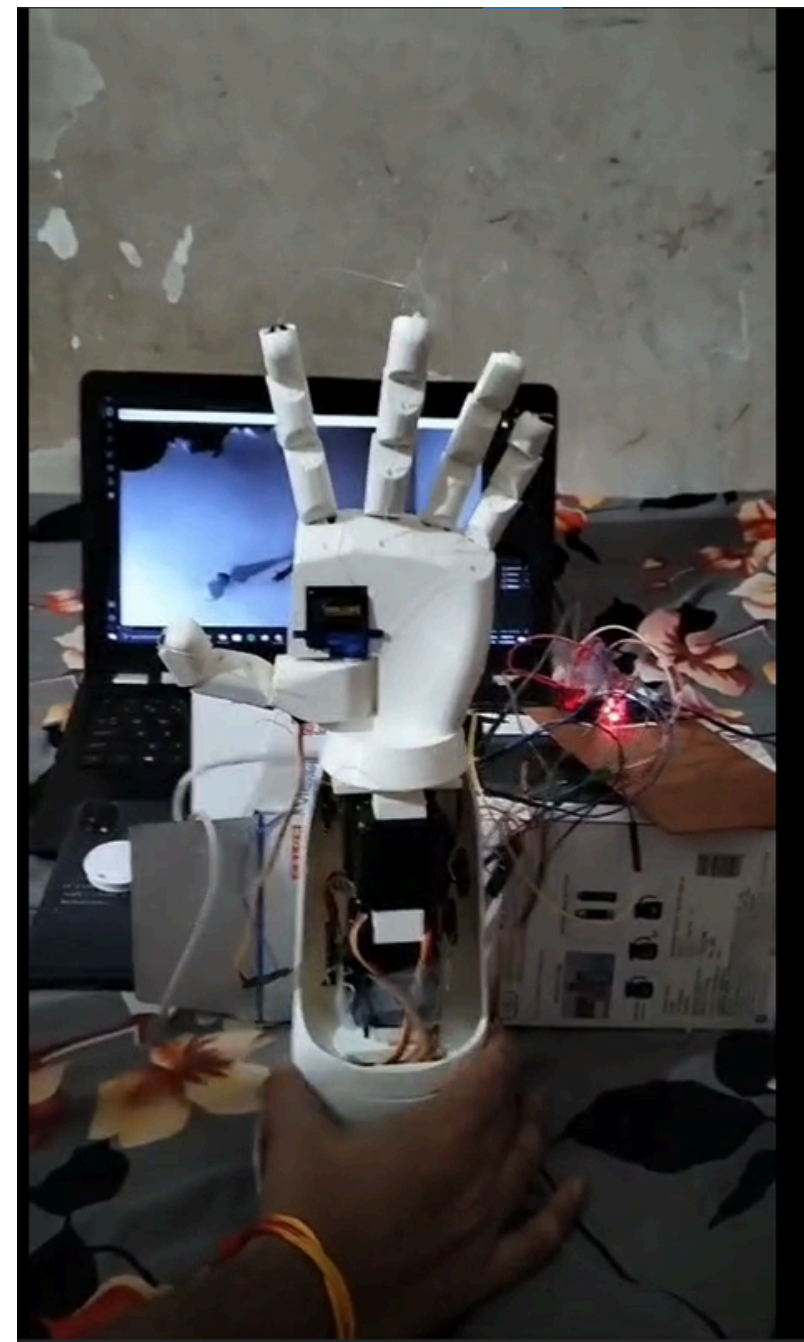
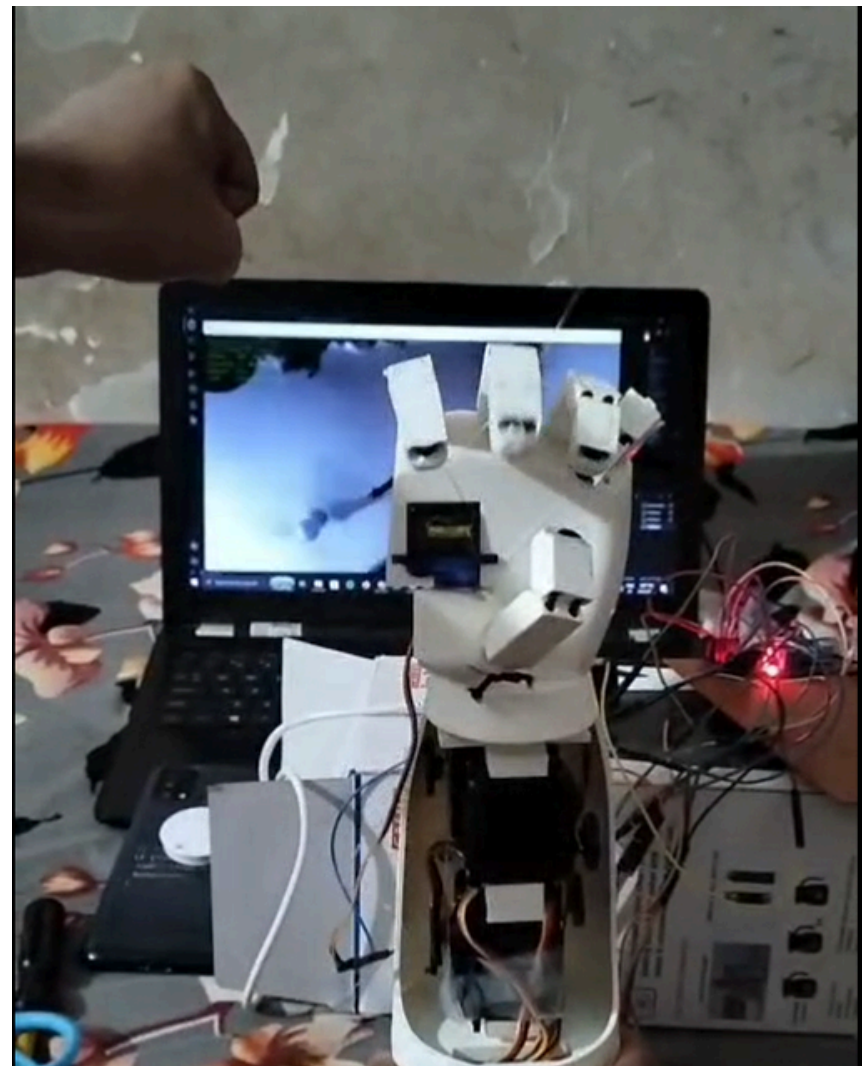
Migrated to two ESP32 boards communicating wirelessly via ESP-NOW protocol, eliminating the need for Bluetooth modules. Flex sensors were retained for the glove unit. Stopped because all flex sensors were found to be damaged — producing only extreme ADC values with no analog variation across the bending range, making finger differentiation completely impossible regardless of software tu

Phase -3 (Final Model):



Eliminated the glove entirely. A USB camera feeds hand landmark data to MediaPipe's pre-trained deep learning model on a host PC, extracting 21 landmarks per frame in real time. A Python script maps landmark positions to 6 servo angles and transmits via UART to a single ESP32, which drives the robotic hand. This gave reliable, contact-free, real-time gesture control with no damaged hardware depend

Phase -3 Additional developments:



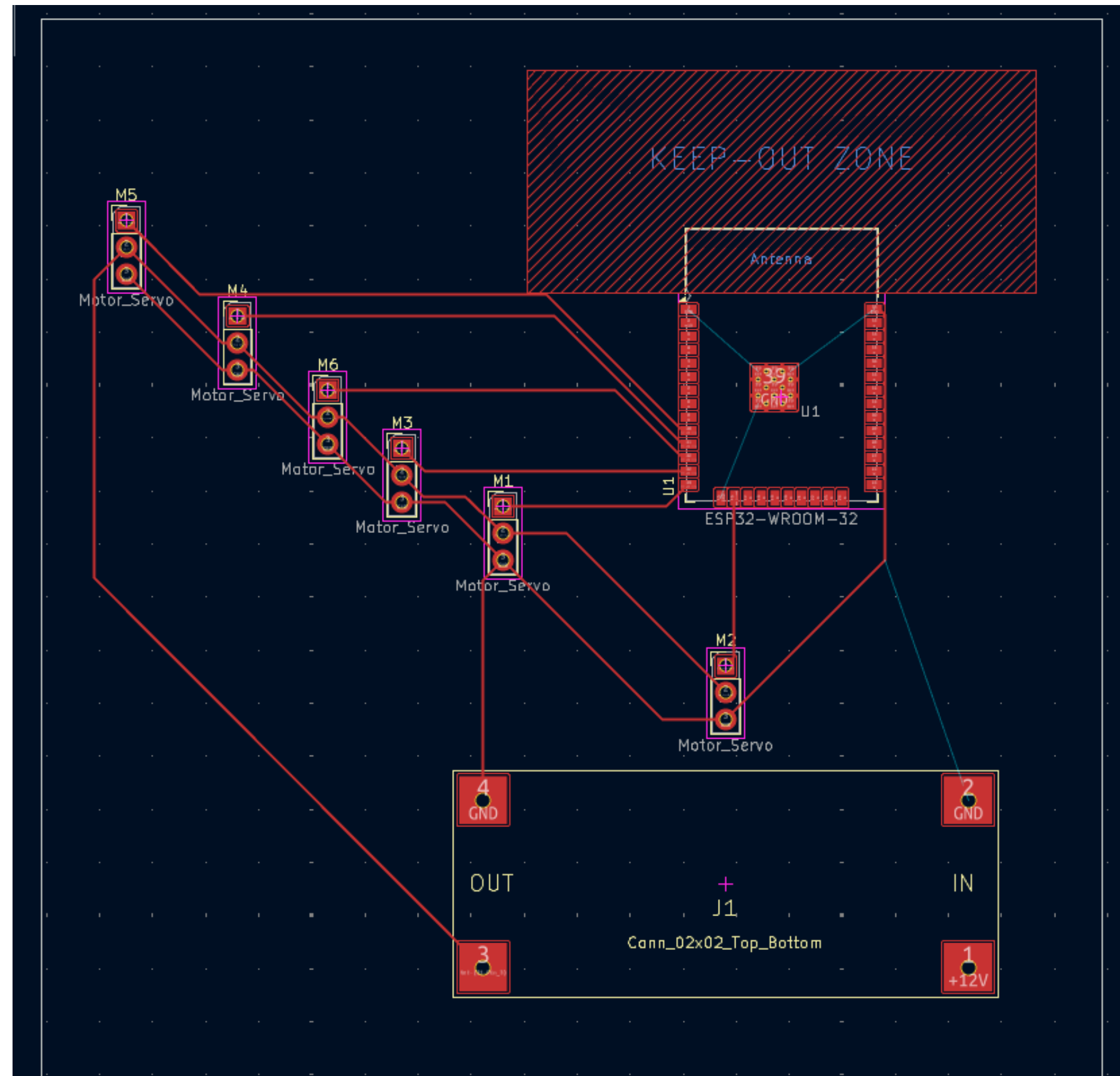
To move beyond hardcoded rule-based finger logic, a custom neural network classifier was trained on self-collected hand landmark data. 200 samples per gesture class were recorded, and a 4-layer dense network was trained achieving 95%+ accuracy. This replaces MediaPipe's landmark rules with a learned model — making the system extensible to any custom gesture set without rewriting logic.

Challenges Faced:



- **Hardware Failures** STM32 Nucleo damaged in lab, Bluepill was a counterfeit — forced complete MCU migration mid-project. No spares available, required full redesign of communication architecture.
- **Sensor Failure** All 5 flex sensors were defective — only outputting rail values at maximum bend with no usable analog range. Discovered only after full glove assembly, requiring complete input mechanism rethink.
- **3D Printed Frame — Support Structure Problem** The robotic hand frame had dense internal support structures that could not be removed with standard tools. Used workarounds including heated needles to melt supports in small holes and screwdrivers with a hammer for larger cavities. Despite this, 2 fingers remain slightly loose due to inaccessible supports affecting joint fit — a mechanical limitation that persists in the final build.
- **Thumb Detection Inversion** MediaPipe labels a physical left hand as 'Right' after horizontal image flip, causing thumb open/close detection to be completely inverted. Required handedness-aware conditional logic to fix — not documented anywhere in MediaPipe's official guides.
- **Wrist Flip Detection** Detecting true hand orientation (palm vs back) from a 2D camera required implementing a cross-product of MCP joint vectors to compute palm normal direction — standard landmark Y comparisons were insufficient.

Bonus (PCB):



Links & References:



Link for all the stl files taken from an open source project.

Link for drive folder containing all the codes as well as block diagram, bill of materials, pcb files and model training results for additional developements.

Link for photos and videos of project and our working time.



Work Distribution



Aditya Gautam

- Worked on designing the circuit throughout different iterations.
- Worked on all the codes of all iterations.
- Worked on PCB design.
- Worked on assembling and customizing the frame.
- Explored programming languages like Embedded C and softwares like Arduino IDE and STM32CubeMX
- Worked on Midsem PPT and Endsem PPT

Ayan Ahmed

- Worked on assembling the frame and customizing it.
- Worked on setting up STM32CubeMX and exploring programming softwares such as Arduino IDE
- Worked on Midsem Evaluation PPT
- Worked on python code.

Abhay

- Worked on designing the circuit throughout different iterations.
- Worked on PCB design.
- Worked on exploring things such as Servo Motors, flex sensors and their functionalities
- Worked on creating a low fidelity prototype with cardboard in order to experiment with it.
- Assisted in component selection based on electrical and mechanical constraints.

Akash

- Worked on assembling the frame and customizing it.
- Worked on exploring things such as Servo Motors, flex sensors and their functionalities
- Worked on creating a low fidelity prototype with cardboard in order to experiment with it.
- Assisted in component selection based on electrical and mechanical constraints.
- Assisted in designing the circuit throughout various iterations.

Our Team:



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Thank You



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